

Tight asymptotic matching cost for one-dimensional fully dynamic spatial matching

ChatGPT (gpt-5.6-sol)*

Partial progress generated on 2026-08-24

Abstract

We present partial progress on an open problem from *Dynamic Spatial Matching*.

1 The open problem

This paper addresses an open problem posed by Yash Kanoria in *Dynamic Spatial Matching* [1] (Economics and Computation (EC), 2022), Section 7.2 (Open directions), page 30 (arXiv v3, Oct 2023). The website catalogue entry is problem 22 (W4285089627_p0) of the [Open Problems in Operations Research](#) collection.

Determine the correct asymptotic scaling (up to constant factors) of the minimum achievable expected average matching cost in the one-dimensional fully dynamic spatial matching model described above as a function of m .

Equivalently, close the logarithmic-factor gap between the existing lower bound $\Omega\left(\frac{\log m}{m}\right)$ and upper bound $O\left(\frac{(\log m)^2}{m}\right)$ on the optimal expected average cost per match (for sufficiently long horizons / in steady state).

The remainder of this document is the partial progress produced by the model, reproduced verbatim from the pipeline output.

Status: partial progress, not a complete solution

Let V_m denote the optimal stationary expected matching cost on $[0, 1]$. The rigorously justified conclusion is

$$\boxed{c \frac{\log m}{m} \leq V_m \leq C \frac{(\log m)^{3/2}}{m}}.$$

The upper bound improves upon $O((\log m)^2/m)$, but the remaining factor of $\sqrt{\log m}$ has not been closed.

*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve $\rightarrow$ A.8 verify $\rightarrow$ A.10 repair write-up; run `run_20260824_071932_24bfd17d, ec-2it-ultra`); two-iteration run, outcome from iteration 2. Independent verifier assessment: REJECTED with progress score 2/3 (major partial progress). Maintained by the AI in Mathematics & Operations Research project.

2 Stationary circle policy and interval transfer

The external result we invoke is the following precise theorem of Coffman–Gilbert–Shor. Fix n points on the unit circle. At every step:

1. draw an independent uniform search point;
2. remove the first stored point encountered in one fixed orientation;
3. insert a new independent uniform point.

The chain has a stationary limiting law, and under that law the mean directed search distance d_n satisfies

$$d_n = \Theta\left(\frac{(\log n)^{3/2}}{n}\right).$$

This is exactly the process described in the [original article](#) and the [Bell Labs record](#); the asymptotic is stated explicitly in [Coffman’s lecture notes](#).

Apply the theorem with $n = m$. A demand D is the fresh uniform search point, the matched supply S is its clockwise successor, that supply is removed, and the replacement A is an independent uniform point. Reversing clockwise and counterclockwise is merely a reflection. Thus every hypothesis of the theorem is met, and for

$$\delta = (S - D) \bmod 1$$

we have

$$\mathbb{E} \delta = O\left(\frac{(\log m)^{3/2}}{m}\right).$$

Now choose an independent Haar-uniform cut Θ and set

$$T_\Theta(x) = (x - \Theta) \bmod 1.$$

Conditional on Θ , all transformed demands and replacements remain mutually independent uniform random variables. Since this conditional product distribution does not depend on Θ , they are also independent uniform random variables unconditionally.

Fix one matched clockwise arc of length δ . The cut lies outside its interior with probability $1 - \delta$, in which case the ordinary interval distance is δ ; it lies inside with probability δ , in which case the interval distance is $1 - \delta$. Therefore

$$\begin{aligned} \mathbb{E}_\Theta [|T_\Theta(S) - T_\Theta(D)| \mid D, S] &= (1 - \delta)\delta + \delta(1 - \delta) \\ &= 2\delta(1 - \delta) \\ &\leq 2\delta. \end{aligned}$$

Consequently, the transformed stationary process is an admissible interval policy, and

$$V_m \leq 2\mathbb{E} \delta = O\left(\frac{(\log m)^{3/2}}{m}\right).$$

Averaging over Θ also shows that some deterministic cut has no larger stationary expectation. The policy itself is simply “take the cyclic successor”; it uses only the current state and the current demand.

Cut coincidences and demand–supply ties have probability zero, and the formula extends continuously to $\delta = 0$.

Combining this with the previously established lower bound, which applies in particular to stationary policies, proves the displayed two-sided estimate.

This argument supplies no mixing-time estimate uniform in m . The fixed- m convergence theorem permits taking the long-horizon limit first, but it does not yield the source paper’s uniform finite-horizon assertions for prescribed $N = N(m)$, arbitrary starts, or well-spread starts.

3 Correct split–merge identities

For the binned clockwise chain, write its cyclic word as K bin symbols and $M \geq 2$ particle symbols. Let

$$a_1, \dots, a_M \geq 0, \quad \sum_{r=1}^M a_r = K,$$

where a_r is the number of bin symbols between the preceding particle and particle r . Particle r is selected with probability a_r/K .

If R denotes the number of bins scanned, then, conditionally on a ,

$$\mathbb{E}[R \mid a] = \frac{1}{K} \sum_{r=1}^M \sum_{j=0}^{a_r-1} j = \frac{1}{2K} \sum_{r=1}^M a_r(a_r - 1).$$

Hence

$$\boxed{\mathbb{E}[R] = \frac{1}{2K} \mathbb{E} \sum_{r=1}^M a_r(a_r - 1).}$$

Deleting particle r merges a_r and a_{r+1} , increasing the quadratic sum by $2a_r a_{r+1}$. Thus the conditional expected increase is

$$\frac{2}{K} \sum_{r=1}^M a_r^2 a_{r+1}.$$

If a post-deletion territory has size s , a uniform insertion splits it at one of its s positions, and its conditional expected contribution to the decrease of the quadratic sum is

$$\frac{s}{K} \cdot \frac{s^2 - 1}{3} = \frac{s^3 - s}{3K}.$$

Writing $\tilde{a}$ for the post-deletion territories and equating the stationary drift gives

$$\boxed{\mathbb{E} \sum_r \tilde{a}_r^3 - K = 6 \mathbb{E} \sum_r a_r^2 a_{r+1}.}$$

These identities are valid, but they leave the adjacent cubic term uncontrolled.

4 Corrected generating-function consequence

Let X_i be the load of bin i , and let L_i be its clockwise scan-territory length, with $L_i = 0$ for an empty bin. Define

$$\beta_n = \mathbb{E} \#\{i : X_i = n\}, \quad \alpha_n = \mathbb{E} \sum_i L_i \mathbf{1}_{\{X_i = n\}},$$

and

$$B(z) = \sum_{n \geq 0} \beta_n z^n, \quad A(z) = \sum_{n \geq 1} \alpha_n z^n.$$

Stationarity gives

$$(K + z - 1) A(z) = zK B(z).$$

Comparing coefficients,

$$(K - 1)\alpha_1 = K\beta_0,$$

and, for $n \geq 2$,

$$(K - 1)\alpha_n + \alpha_{n-1} = K\beta_{n-1}.$$

Because $L_i \geq 1$ whenever $X_i > 0$, we have $\alpha_n \geq \beta_n$. Therefore, for $n \geq 2$,

$$K\beta_{n-1} \geq (K - 1)\beta_n + \beta_{n-1},$$

and hence

$$\beta_n \leq \beta_{n-1} \quad (n \geq 2).$$

The missing first comparison is only

$$\beta_1 \leq \frac{K}{K - 1} \beta_0,$$

not $\beta_1 \leq \beta_0$.

The failure is exact already for $K = M = 2$. For the ordered states

$$(2, 0), \quad (1, 1), \quad (0, 2),$$

the transition matrix is

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix},$$

whose stationary distribution is

$$\left(\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right).$$

Thus

$$(\beta_0, \beta_1, \beta_2) = \left(\frac{1}{2}, 1, \frac{1}{2}\right).$$

Hence the former assertion of monotonicity starting at $n = 0$ is false; the fully repaired statement is only

$$\beta_1 \geq \beta_2 \geq \beta_3 \geq \cdots.$$

5 The bounded-hazard route and its exact gap

Suppose a policy satisfies the pointwise conditional bound

$$\Pr(\text{supply } i \text{ is deleted} \mid \text{full current state}) \leq \frac{\gamma}{M}$$

for every supply i . Let B have insertion probability p , let Y be the number of supplies in B , and let Λ_B be the conditional probability that the deletion occurs in B . Then

$$\Lambda_B \leq \frac{\gamma Y}{M}.$$

Writing $\pi_n = \Pr(Y = n)$, the stationary flow across level $n + \frac{1}{2}$ gives

$$p \mathbb{E}[\mathbf{1}_{\{Y=n\}}(1 - \Lambda_B)] = (1 - p) \mathbb{E}[\mathbf{1}_{\{Y=n+1\}}\Lambda_B].$$

For $n \leq M/(2\gamma)$,

$$\frac{p}{2} \pi_n \leq \frac{\gamma(n+1)}{M} \pi_{n+1},$$

so

$$\pi_{n+1} \geq \frac{Mp}{2\gamma(n+1)} \pi_n.$$

With $L = \lfloor Mp/(4\gamma) \rfloor$, the ratio is at least 2 for $0 \leq n < L$, whence

$$\boxed{\Pr(Y = 0) \leq 2^{-L}.$$

Also,

$$\boxed{\Pr(Y = 1 \text{ and its unique supply is deleted}) = \mathbb{E}[\mathbf{1}_{\{Y=1\}}\Lambda_B] \leq \frac{\gamma}{M}.$$

The premise is genuinely pointwise. For the clockwise chain, a particle with territory a_r has conditional deletion probability a_r/K , which can be macroscopic. Exchangeability—or merely an unconditional mean hazard of $1/M$ —does not imply the required hypothesis.

Moreover, empty-block control alone does not provide a cheap capacitated transport. For example, fix $\gamma \geq 1$ and place approximately $m/(4\gamma)$ evenly spaced supplies in $[0, \frac{1}{2}]$, with the remaining supplies evenly spaced in $(\frac{1}{2}, 1]$. Every geometric gap is $O(\gamma/m)$. Nevertheless, if every supply may receive selection mass at most γ/m , then the total selectable mass in the left half is at most $1/4$. Therefore at least $1/4$ of the demand mass in $[0, \frac{1}{2}]$ must be sent across $1/2$. Even in the cheapest arrangement,

$$\text{cost} \geq \int_{1/4}^{1/2} \left(\frac{1}{2} - d \right) dd = \frac{1}{32}.$$

Thus small empty gaps do not exclude positive-density Hall deficits.

For the hazard route, the exact missing construction is a state-dependent kernel $q_x(i \mid d)$ satisfying

$$\lambda_i(x) = \int_0^1 q_x(i \mid d) dd \leq \frac{\gamma}{m}$$

and possessing a stationary law Π_m for which

$$\mathbb{E}_{\Pi_m} \int_0^1 \sum_i q_x(i \mid d) |x_i - d| \, dd = O\left(\frac{\log m}{m}\right).$$

No such construction, nor an equivalent positive-density discrepancy estimate, is proved here.

Hence the monotonicity error is fully repaired, but the fundamental asymptotic question remains open in this attempt. The strongest proved result is exactly

$$\Omega\left(\frac{\log m}{m}\right) \leq V_m \leq O\left(\frac{(\log m)^{3/2}}{m}\right).$$

References

- [1] Yash Kanoria. *Dynamic Spatial Matching*. Economics and Computation (EC), 2022. <https://doi.org/10.1145/3490486.3538278>